

Model

Evaluations

Evaluations:

1. ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
2. BLEU (Bilingual Evaluation Understudy)
3. METEOR
4. BERTScore

1. ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

Measures overlap of n-grams between generated and reference summaries.

- ROUGE-1: Overlap of unigrams (words)
- ROUGE-2: Overlap of bigrams (pairs of words)
- ROUGE-L: Longest common subsequence (captures fluency)

Higher is better.

Focuses on how much of the reference is covered by the prediction.

2. BLEU (Bilingual Evaluation Understudy)

Measures precision of n-grams (how many predicted n-grams exist in the reference).

Common in machine translation, also used in summarization.

Higher is better.

Focuses on how much of the reference is covered by the prediction.

3. METEOR

Considers synonyms, stemming, and word order.

More semantically aware than BLEU.

Balances precision and recall.

Better aligned with human judgment than BLEU.

4. BERTScore

Uses BERT embeddings to compare similarity at the semantic level.

Outputs Precision, Recall, and F1 based on contextual token similarity.

Captures meaning even if wording is different.

Most suitable for abstractive summarization.

Evaluations

Metric	Score	Interpretation
ROUGE-1	0.3609	Unigram (word-level) overlap with the reference summary. Shows that the summary picks up important terms.
ROUGE-2	0.1838	Bigram (two-word phrase) overlap, suggesting more deviation in phrasing compared to ground truth.
ROUGE-L	0.2019	Overlap in longest common subsequence, indicating structural differences in summary flow/order.

Evaluations

The model captures key terms, but the phrasing and structure of summaries often differ from the references.

Which is normal in abstractive summarization, especially for complex domains like legal texts.

Evaluations

Metric	Score	Interpretation
BLEU	0.0072	This metric expects almost exact word overlap. Since BART is abstractive, it generates paraphrased outputs, which BLEU penalizes.
METEOR	0.1676	Captures semantic similarity and synonym matching better than BLEU. Score indicates partial similarity.

Evaluations

BLEU is not well-suited and often underestimates quality in abstractive generation. Therefore, we can safely deprioritize BLEU for legal summarization.

METEOR suggests that the model captures some semantic essence, and finer or case-specific details.

Evaluations

Metric	Score	Interpretation
BERTScore Precision	0.8548	The generated summary is highly relevant and aligned with the reference content.
BERTScore Recall	0.8326	The summary covers most of the important information from the reference.
BERTScore F1	0.8435	Strong semantic similarity — excellent performance for abstractive legal summarization.

Evaluations

Despite low lexical overlap (ROUGE/BLEU), the summaries are meaningfully accurate.

BERTScore confirms the model is legally and contextually faithful which is a strong sign of real-world utility.

Evaluations

Lexical metrics like (ROUGE/BLEU) are moderate/low which is expected in abstractive summarization.

BERTScore is excellent → Model understands and reproduces the meaning of legal cases well.

Low BLEU is not concerning in our case.

Overall, the fine-tuned model performs very well semantically, which is crucial for legal AI applications.